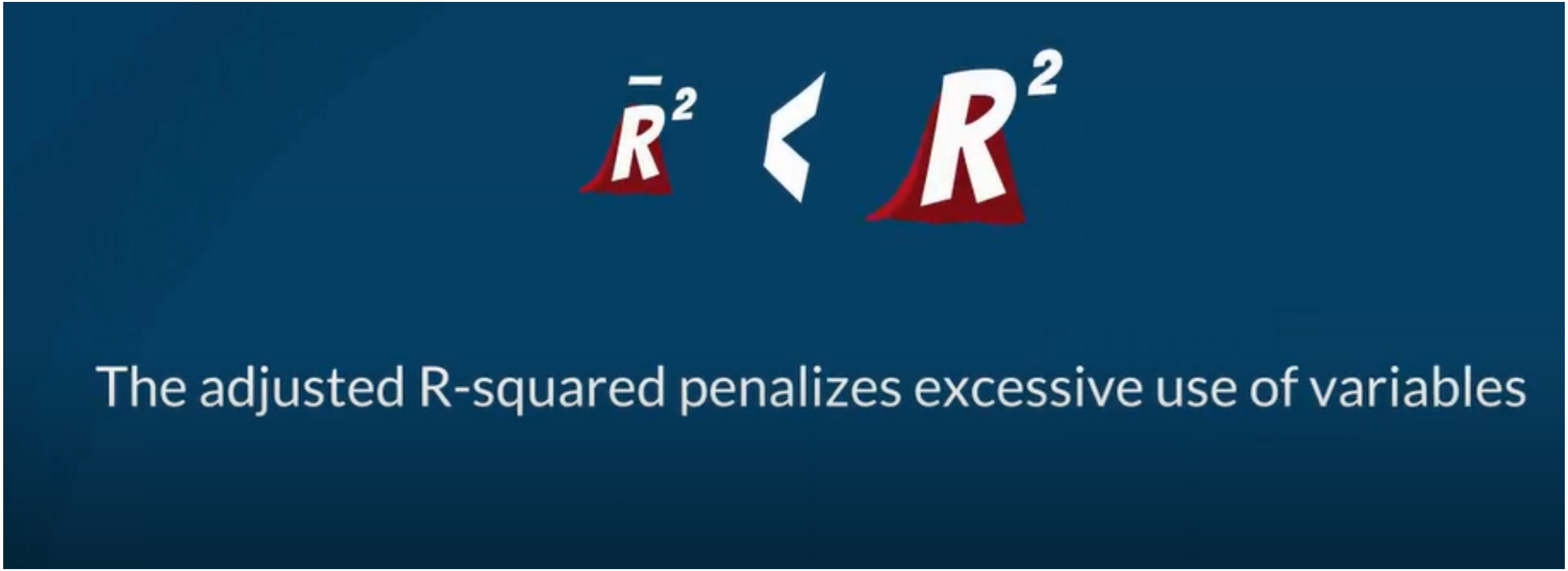


Adjusted R-squared

1. The R-squared measures how much of the total variability is explained by our model

2. Multivariate regressions are always better than univariate, as with each additional variable you add, the explanatory power may only increase or stay the same


$\bar{R}^2 < R^2$
The adjusted R-squared penalizes excessive use of variables

Adjusted R-squared

GPA	SAT	Random
2.40	1714	2
2.52	1664	2
2.54	1760	1
2.74	1685	3
2.83	1693	1
2.91	1670	2
3.00	1764	2
3.00	1764	1
3.01	1792	1
3.01	1850	1
3.02	1735	3
3.07	1775	2
3.08	1735	2
3.08	1712	2
3.12	1773	1
3.17	1872	2
3.17	1755	3
3.17	1674	3
3.17	1842	1
3.19	1786	3
3.19	1761	1
3.19	1722	2
3.20	1663	2
3.21	1687	1
3.24	1974	3
3.28	1826	3
3.28	1787	3

A variable that assigns 1, 2 or 3 randomly to each student

Adjusted R-squared

GPA	SAT	Random
2.40	1714	2
2.52	1664	2
2.54	1760	1
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3.08	1712	2
3.12	1773	1
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3.17	1674	3
3.17	1842	1
3.19	1786	3
3.19	1761	1
3.19	1722	2
3.20	1663	2
3.21	1687	1
3.24	1974	3
3.28	1826	3

$$\hat{y} = b_0 + b_1 x_1 + b_2 x_2$$

$$\text{GPA} = b_0 + b_1 SAT + b_2 Rand123$$

Adjusted R-squared

$$\text{GPA} = b_0 + b_1 SAT + b_2 Rand123$$

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.638
R Square	0.407
Adjusted R Square	0.393
Standard Error	0.212
Observations	100

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	2	2.494	1.247	27.830	0.000
Residual	97	3.629	0.045		
Total	99	6.123			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
Intercept	0.247	0.416	0.592	0.556	-0.582	1.075	-0.582	1.075
SAT score	0.002	0.000	7.458	0.000	0.001	0.002	0.001	0.002
Rand 1,2,3	0.012	0.028	0.418	0.677	-0.044	0.068	-0.044	0.068

Adjusted R-squared

$$\text{GPA} = b_0 + b_1 SAT + b_2 Rand123$$

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We were penalized for adding a variable that had no explanatory power

Adjusted R-squared

SUMMARY OUTPUT

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You should exclude useless information



Basis for comparing models

1. Same dependent variable (y)

2. Same dataset